

Beyond Class Boundaries: Federated Visual Primitive Sharing with Text-Guided Adaptation

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Abstract

Personalized Federated Learning (pFL) effectively addresses the challenge of statistical heterogeneity in traditional Federated Learning (FL), with feature alignment methods (e.g., FedProto) standing out due to their communication efficiency and model-agnostic design, making them practically viable in real-world non-IID scenarios. These methods directly align class-level features across clients without requiring model parameter transmission. However, they represent each class as a holistic prototype, which limits the diversity and expressiveness of shared features. This restriction hampers the model's ability to generalize across clients and impedes personalized adaptation, as clients lack sufficient semantic components to reconstruct discriminative features tailored to their local data distributions. To overcome these limitations, we propose Federated Visual Primitive Learning (FedVPL), a novel framework comprising two key components: (1) **Visual Primitive Space Sharing**, which decomposes class-level features into semantically meaningful and reusable visual primitives, enabling cross-client and cross-class sharing to enrich feature diversity and decouple communication cost from the number of classes, significantly improving efficiency; and (2) **Text-Guided Semantic Alignment**, a parameter-free personalization mechanism that leverages external language priors to align shared primitives with client-specific semantics, without requiring additional communication overhead. Extensive experiments across diverse non-IID benchmarks demonstrate that FedVPL

substantially outperforms state-of-the-art baselines, achieving up to a 5.62% improvement in accuracy, reducing communication overhead by at least 12.5×, and effectively addressing generalization and personalization challenges in heterogeneous federated environments.

CCS Concepts

• **Computing methodologies** → **Machine learning**: *Distributed algorithms*; • **Human-centered computing** → **Ubiquitous and mobile computing**.

Keywords

Personalized Federated Learning, Knowledge Sharing, Feature Representations

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1 Introduction

Mobile and Web-of-Things (WoT) devices at the network edge generate vast amounts of data daily [7, 36]. However, due to privacy concerns, this data cannot be directly accessed by data-driven models [3, 6, 8, 12, 16, 23, 34] that require large volumes of data for optimal performance. Federated Learning (FL) has emerged as a promising paradigm that enables collaborative model training across distributed clients while preserving data privacy [25]. In FL, each client trains a local model on its private data and shares only model updates or features with a central server for aggregation, which mitigates privacy risks while leveraging collective knowledge from diverse parties, making it particularly attractive for web- and edge-centered applications such as mobile sensing, healthcare, and edge intelligence [11, 26, 30]. However, traditional

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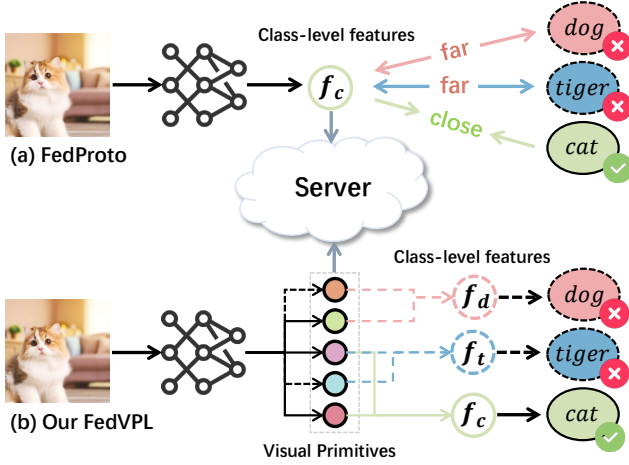


Figure 1: Comparison of (a) class-level feature sharing and our proposed (b) visual primitive space sharing.

FL algorithms, such as FedAvg [25] assume that data distributions across clients are independent and identically distributed (IID), an assumption often violated in real-world scenarios. Edge and WoT devices typically collect user- or context-specific data, producing strongly non-IID distributions [15, 40, 44]. Under such heterogeneity, a single global model often fails to generalize, because a single parameter set cannot reconcile the conflicting feature statistics and objectives present across clients.

To address this challenge, pFL has been proposed to tailor models to individual client distributions. Existing pFL approaches can be broadly categorized into four paradigms: (1) *Model-splitting strategies* [2, 13, 21, 42] decouple model into global and local components for parameter sharing and personalized learning, respectively; (2) *Regularization-based techniques* [1, 19, 20, 31] balance global and local objectives via client-specific constraints; (3) *Knowledge distillation frameworks* [14, 29, 37] transfer knowledge through lightweight student models, and (4) *Feature alignment methods* [32, 38, 47] align class-level semantic vector across clients. While the first three paradigms offer strong personalization, they typically incur high communication cost and introduce architectural constraints that limit deployment scalability. In contrast, feature alignment approaches are communication-efficient and model-agnostic, making them practical for large-scale heterogeneous settings. Despite their advantages, existing feature alignment methods suffer from a key limitation: they represent each class as a holistic prototype, enforcing a rigid one-to-one mapping between semantic categories and global features. This severely limits the expressiveness and diversity of shared representations, particularly when clients exhibit varying label spaces, class imbalance, or intra-class variation. Moreover, methods like FedProto [32] are sensitive to the quality of global prototypes; noisy or biased prototypes can propagate errors during local adaptation [45].

In this work, we address the representational bottlenecks of class-level alignment by introducing a more compositional and flexible semantic structure. Specifically, we propose FedVPL, a novel framework that replaces rigid class-level prototypes with a shared

space of *visual primitives*—semantically meaningful, reusable feature units that can be composed into diverse high-level representations. As illustrated in Figure 1, instead of aligning entire class features, FedVPL aligns and shares primitive bases across clients. This leads to three key advantages: (1) Composability of primitives exponentially increases the diversity of semantic representations, supporting finer-grained personalization; (2) Primitive bases are low-dimensional and class-agnostic, enabling constant communication cost regardless of label space; (3) Cross-client alignment at the primitive level improves both generalization and adaptability.

To implement this idea, FedVPL comprises two key components: (1) **Visual Primitive Space Sharing**, which learns and maintains a global space of aligned primitive bases shared across clients and classes; and (2) **Text-Guided Semantic Alignment**, a parameter-free personalization mechanism that leverages external language priors to adapt shared primitives to client-specific semantics without requiring additional communication overhead. To sum up, our contributions are summarized as follows:

- We propose FedVPL, the first federated learning framework that replaces class-level feature sharing with visual primitive space sharing, significantly enhancing the diversity and flexibility of cross-client representations.
- We introduce a text-guided semantic alignment mechanism that enables personalized adaptation by aligning visual primitives with client-specific semantics in a parameter-free and communication-efficient manner.
- Extensive experiments on diverse non-IID benchmarks demonstrate that FedVPL outperforms state-of-the-art pFL baselines by up to 5.62% in accuracy, while reducing communication overhead by at least 12.5× compared to existing feature alignment methods.

2 Related Work

Personalized Federated Learning (pFL) aims to tackle statistical heterogeneity by enabling client-specific models while leveraging collaborative training. Existing approaches can be broadly categorized into four paradigms, each with unique mechanisms and trade-offs.

Regularization-based methods [19, 20, 31] enforce consistency by penalizing deviations between local and global models. For example, FedProx [20] introduces a proximal term to stabilize local updates, while Ditto [19] and pFedMe [31] optimize personalized models to balance global guidance with local adaptation. FedDyn [1] adjusts regularization strength dynamically based on client-specific data distributions. Although these methods mitigate client drift, their focus on parameter-level alignment overlooks feature space diversity, limiting their capacity to resolve conflicting semantic representations across heterogeneous clients.

Model-splitting strategies [4, 13, 22, 39, 41–43, 46] architecturally separate shared and personalized components. FedPer [2] and LG-FedAvg [21] isolate global feature extractors from client-specific classifiers, whereas FedRep [5] decouples training phases for these components. Advanced variants like FedRoD [4] employ dual-head architectures to disentangle global and personalized classification tasks. While effective, such rigid architectural partitioning

increases communication overhead for transmitting partial parameters and imposes synchronization constraints during federated training.

Knowledge distillation frameworks [14, 29, 37] transfer knowledge through lightweight models or soft targets. FedKD [37] compresses global knowledge into compact student models, reducing communication costs by sharing only distilled outputs. FML [29] enables mutual knowledge transfer between global and personalized models. These approaches are suitable for heterogeneous model architectures but often suffer from overly abstract representations; distilled outputs may oversimplify complex local semantics and fail to preserve discriminative features critical for personalization. Moreover, unlike feature-level methods, distillation indirectly mimics teacher behavior rather than enforcing geometric consistency in the representation space.

Feature alignment methods [32, 38, 47] directly regulate the semantic structure of learned representations. FedProto [32] pioneers class-level prototype sharing, where clients transmit only feature centroids for server aggregation, offering communication efficiency and model-agnostic deployment. Subsequent variants, such as FedPCL [33] and FedPAC [38], extend this idea with contrastive learning and global semantic alignment. Despite their success, these methods typically enforce a fixed, one-to-one mapping between classes and prototypes, which restricts representational flexibility. For instance, FedProto’s reliance on class-specific prototypes assumes uniform semantic granularity across clients, leading to biased representations when local class distributions diverge [45].

3 Preliminaries

3.1 Client Updating

In a standard cross-device personalized federated learning of web scenario with N clients, each client i trains a personalized model $\mathbf{W}^i$ on its private dataset $\mathcal{D}^i = \{(x_j, y_j)\}_{j=1}^{|\mathcal{D}^i|}$, where $|\mathcal{D}^i|$ denotes the dataset size. To balance shared knowledge and local adaptation, model parameters are typically partitioned into: (1) global components (e.g., feature extractor f_θ) shared across clients to capture universal patterns; (2) local components (e.g., classifier head $g_{\mathbf{w}^i}$) tailored to client-specific tasks. During local training, client i minimizes the empirical risk:

$$\arg \min_{\theta, \mathbf{w}^i} \sum_{(x, y) \in \mathcal{D}^i} \mathcal{L}_s(g_{\mathbf{w}^i}(f_\theta(x)), y). \quad (1)$$

Our proposed FedVPL departs from this paradigm by introducing a discrete and composable visual primitive space during client updates, which reduces the dependency on explicit classifier head and facilitates richer representation sharing across heterogeneous clients even before server aggregation.

3.2 Server Aggregation

The server aggregates client updates to refine shared knowledge while accommodating heterogeneity. There are two main strategies:

3.2.1 Model Parameter Aggregation. In this FL paradigm, the server averages local parameters θ^i from N clients at communication

round t via weighted averaging:

$$\theta^{\text{global}} = \frac{1}{N} \sum_{i=1}^N \theta_t^i. \quad (2)$$

However, due to heterogeneity, divergent optimization directions across clients may cause gradient conflict during aggregation, leading to client model drift. Additionally, if the model contains numerous parameters, large updates cause communication inefficiency.

3.2.2 Feature Aggregation. Class-level feature alignment methods, such as FedProto [32], aggregate feature prototypes instead of parameters. For each class c , client i computes a prototype $\mathbf{a}_c^i$, defined as the mean feature vector of samples in $\mathcal{D}^i$ that belong to class c . The server then aggregates class-wise prototypes from clients:

$$\mathbf{a}_c^{\text{global}} = \frac{1}{N_c} \sum_{i=1}^{N_c} \mathbf{a}_c^i, \quad (3)$$

where N_c denotes the number of clients containing class c . The resulting global prototypes are redistributed to clients to guide local feature alignment.

Our proposed FedVPL also belongs to the family of feature aggregation approaches. However, instead of class-level prototype sharing, it facilitates visual primitive space sharing, which enhances the diversity of feature representations and communication efficiency across clients.

4 Method

Our goal is to collaboratively train a representation model that supports both global knowledge sharing and local personalization, under constraints of limited communication and privacy preservation. We achieve this goal by two key innovations: (1) Visual Primitive Space Sharing, and (2) Text-Guided Semantic Alignment. This section details our proposed modules.

4.1 Visual Primitive Space Sharing

As illustrated in Figure 2(a), each client maintains a local visual primitive space $\mathcal{S}_{\text{local}} \in \mathbb{R}^{K \times d_v}$ initialized from the server’s global visual primitive space $\mathcal{S}_{\text{global}} \in \mathbb{R}^{K \times d_v}$, where K denotes the number of primitive bases in the space and d_v is the dimension of the primitive base. During local training, the image encoder f_θ extracts latent features $z = f_\theta(x)$, which are then decomposed into local visual primitives through a lightweight visual pattern projector $p_{\mu, \Sigma}$ as follows:

$$\mathbf{v}_\mu, \mathbf{v}_\Sigma = p_{\mu, \Sigma}(z), \quad (4)$$

where $\mathbf{v}_\mu \in \mathbb{R}^{M \times d_v}$ captures core visual patterns and $\mathbf{v}_\Sigma \in \mathbb{R}^{M \times d_v}$ models client-specific variations, M is the number of decomposed visual primitives of the input image.

To increase representation diversity, we draw inspiration from vector quantization techniques [9, 35] to discretize the latent visual space of atomic visual primitives that transcend class boundaries. The quantization mechanism operates via nearest-neighbor retrieval in the visual primitive space $\mathcal{S}_{\text{local}} \in \mathbb{R}^{K \times d_v}$: for each core visual pattern $\mathbf{v}_\mu^m \in \mathbf{v}_\mu$, the closest primitive $\mathbf{p}^*$ is identified through Euclidean distance minimization:

$$\mathbf{p}^* = \arg \min_{\mathbf{p}_k \in \mathcal{S}_{\text{local}}} \|\mathbf{v}_\mu^m - \mathbf{p}_k\|_2, \quad (5)$$

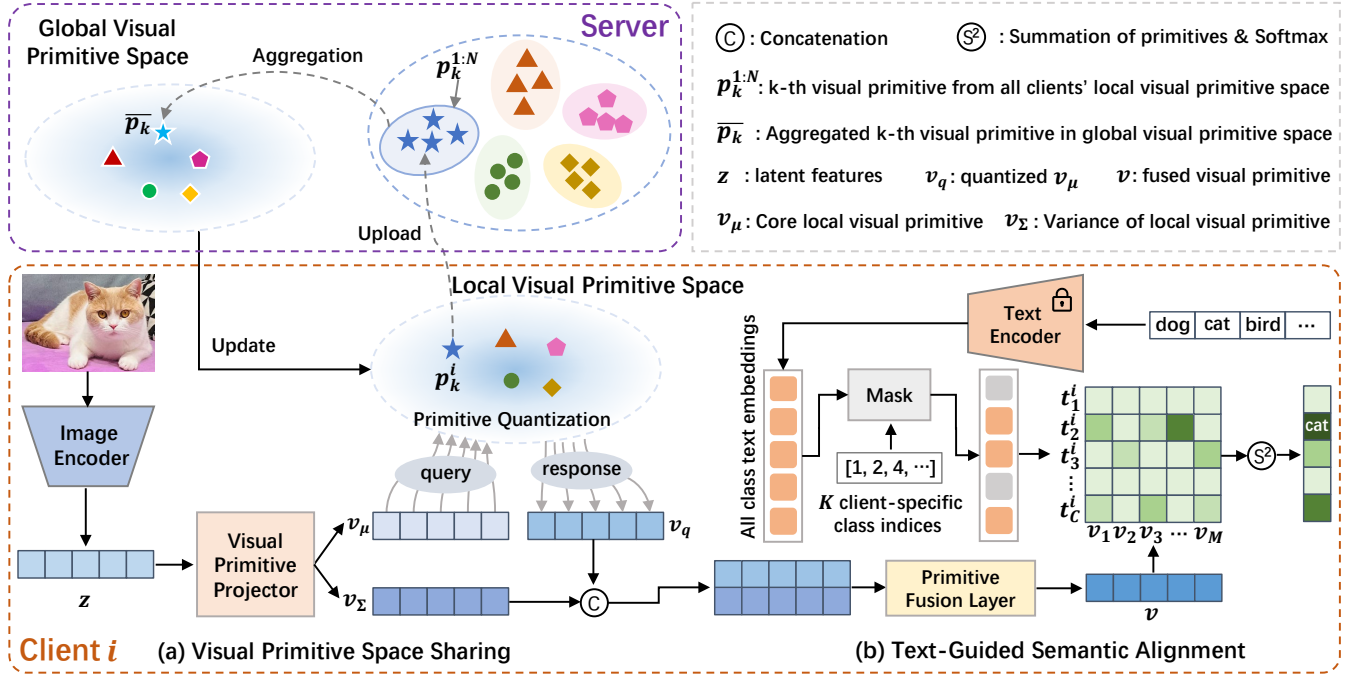


Figure 2: Overview of our proposed Federated Visual Primitive Learning (FedVPL).

and replacing the original core visual patterns $\mathbf{v}_\mu$ with the quantized primitives $\mathbf{v}_q \in \mathbb{R}^{M \times d_v}$. To ensure the primitive space reflects meaningful and recurring visual patterns, we optimize the encoder and the local primitive space jointly with:

$$\mathcal{L}_{\text{space}} = \|\text{sg}[\mathbf{v}_\mu] - \mathbf{v}_q\|_2^2 + \lambda \|\mathbf{v}_\mu - \text{sg}[\mathbf{v}_q]\|_2^2, \quad (6)$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operation, and λ is the balancing factor. The first term updates the encoder to align features with the current primitive space, while the second term refines the primitives, pulling them toward high-density regions of the feature distribution. This dual optimization ensures that the primitive space $\mathcal{S}_{\text{local}}$ dynamically captures frequent and discriminative visual patterns.

To integrate client-specific variations, the quantized primitives $\mathbf{v}_q \in \mathbb{R}^{M \times d_v}$ are concatenated with the corresponding variance component $\mathbf{v}_\Sigma \in \mathbb{R}^{M \times d_v}$ and undergo dimensional adjustment via a lightweight primitive fusion layer ϕ as follows:

$$\mathbf{v} = \phi(\mathbf{v}_q \oplus \mathbf{v}_\Sigma) \in \mathbb{R}^{M \times d_t}, \quad (7)$$

where $\oplus$ represents feature concatenation, and d_t is the dimension of text embeddings in downstream alignment.

During server-side aggregation, primitive bases in the clients' local space are merged via weighted averaging based on client data volumes:

$$\mathcal{S}_{\text{global}} = \sum_{i=1}^N \frac{|\mathcal{D}^i|}{\sum_{j=1}^N |\mathcal{D}^j|} \mathcal{S}_{\text{local}}^i. \quad (8)$$

After aggregation, the global visual primitive space is redistributed to all participated clients to initialize subsequent local training rounds, completing a federated cycle. This process consolidates cross-client visual knowledge into a globally consistent primitive

space, where each primitive serves as a reusable "building block" for downstream tasks.

By grounding the federated learning process in a discrete, composable primitive space, our framework transcends the limitations of class-centric feature aggregation, enabling robust generalization in heterogeneous environments. The interplay between local primitive refinement and global aggregation establishes a cohesive knowledge-sharing paradigm, where visual primitives act as the universal "language" for cross-client collaboration.

4.2 Text-Guided Semantic Alignment

To enable personalization, we propose a parameter-free alignment mechanism that bridges visual primitives and language priors, as shown in Figure 2(b), the process involves two key steps:

4.2.1 Semantic Anchoring with Frozen Language Models. A pre-trained text encoder (CLIP [28]) encodes all class names into a global semantic matrix $\mathbf{T} \in \mathbb{R}^{C \times d_t}$, where C is the total class count and d_t is the text embedding dimension. To adapt to client-specific label distributions, for each client i , a binary mask $\mathbf{M}^i \in \{0, 1\}^C$ filters $\mathbf{T}$ to retain only local class indices I^i :

$$\mathbf{t}_c^i = m_c \cdot \mathbf{t}_c, \quad m_c = \begin{cases} 1 & \text{if } c \in I^i \\ 0 & \text{otherwise} \end{cases}, \quad (9)$$

where $\mathbf{T}^i = \{\mathbf{t}_1^i, \mathbf{t}_2^i, \dots, \mathbf{t}_C^i\} \in \mathbb{R}^{C \times d_t}$ represents the client's filtered semantic space. This masking operation projects the global semantic matrix onto the client's local label subset without introducing learnable parameters.

4.2.2 Cross-Modal Primitive-Text Alignment. Each visual primitive $\mathbf{v}_m$ and text embedding $\mathbf{t}_c^i$ is ℓ_2 normalized for stable similarity computation:

$$\hat{\mathbf{v}}_m = \frac{\mathbf{v}_m}{\|\mathbf{v}_m\|_2 + \epsilon}, \quad \hat{\mathbf{t}}_c^i = \frac{\mathbf{t}_c^i}{\|\mathbf{t}_c^i\|_2 + \epsilon}, \quad (10)$$

where $\epsilon = 10^{-9}$ prevents numerical instability.

The model computes cosine similarities between all M visual primitives and C masked text descriptors:

$$\mathbf{S}_{m,c} = \hat{\mathbf{v}}_m \cdot \hat{\mathbf{t}}_c^{i^\top} \in \mathbb{R}^C, \quad (11)$$

where $\mathbf{S}_{m,c}$ serves as the primitive-level classification evidence at class c , which is derived by calculating the cosine similarity of the m -th visual primitive and the c -th text embedding. Furthermore, all M primitive-level evidences are aggregated via summation to form class-wise scores:

$$\mathbf{S} = \sum_{m=1}^M \mathbf{S}_{m,c} \in \mathbb{R}^C. \quad (12)$$

A temperature-scaled softmax is then applied to generate final probabilities:

$$P(y = c|x) = \frac{\exp(\mathbf{S}_c/\tau)}{\sum_{c \in \mathcal{I}^i} \exp(\mathbf{S}_c/\tau)}, \quad (13)$$

where τ controls prediction confidence. The classification loss optimizes cross-entropy over local labels as follows:

$$\mathcal{L}_{\text{cls}} = - \sum_{(x,y) \in \mathcal{D}^i} \sum_{c \in \mathcal{I}^i} \mathbb{I}(y = c) \log P(y = c|x), \quad (14)$$

where $\mathbb{I}(y = c)$ is an indicator function.

Without relying on a parameterized classifier head, this mechanism aligns learned visual primitives with client-specific semantics through frozen language models. The resulting classification process is interpretable, communication-free, and inherently personalized via local label masking.

Finally, the overall optimization objective is as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{space}} + \mathcal{L}_{\text{cls}}. \quad (15)$$

And the learning procedure of FedVPL is shown in Appendix A.

5 Experiments

5.1 Statistically Heterogeneous Settings

We evaluate the robustness of the compared federated learning methods under three statistically heterogeneous settings: pathological, practical, and real-world. Each setting represents varying degrees and forms of non-IID data distributions. Figure 3 illustrates the data distribution on Cifar10 and Cifar100 datasets [17] in the pathological and practical settings with 20 clients.

5.1.1 Pathological Setting. The pathological setting simulates extreme heterogeneity by assigning clients data from a small, non-overlapping subset of classes, creating highly fragmented distributions. We utilize the MNIST [18] and CIFAR10 [17] datasets to simulate this setting. Following FedAvg [25], each client is randomly assigned data from two out of ten classes, ensuring no label overlap among clients. This scenario is designed to stress-test the model's ability to generalize in environments where data diversity is severely limited for each client.

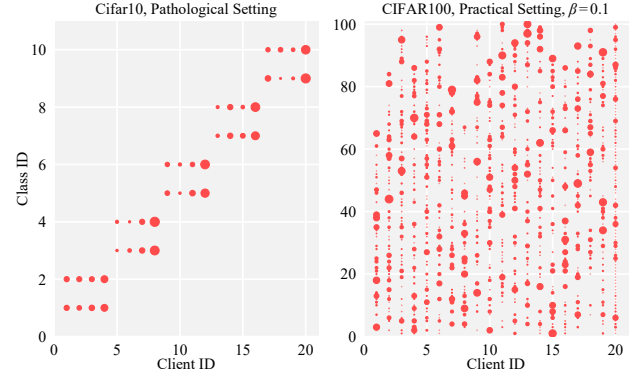


Figure 3: Data distribution on Cifar10 and Cifar100 datasets. The size of a circle indicates the number of local data.

5.1.2 Practical Setting. The practical setting simulates more realistic heterogeneity by using a Dirichlet distribution to control the concentration of class distributions across clients, as commonly done in prior works [39, 44, 45]. We apply this setting to the CIFAR100 [17] dataset by sampling proportions $q_{c,i} \sim \text{Dir}(\beta)$, where $q_{c,i}$ denotes the fraction of samples from class c assigned to client i . We evaluate three levels of heterogeneity with $\beta \in \{0.1, 0.3, 0.5\}$, which represent increasingly balanced distributions.

5.1.3 Real-world Setting. To simulate heterogeneity arising from real-world scenarios, we employ two domain adaptation datasets: OfficeCaltech10 [10] and DomainNet [27]. The OfficeCaltech10 dataset consists of images from four distinct domains (three from Office-31 and one from Caltech-256), each representing different acquisition conditions and background styles. DomainNet, on the other hand, contains images from six different sources: Clipart, Infograph, Painting, Quickdraw, Real, and Sketch. In both datasets, each client is assigned data from only one domain, which introduces natural non-IIDness due to domain-specific visual characteristics and labeling biases. These settings reflect the challenges of real federated web environments, where data heterogeneity arises from user behavior or hardware differences.

5.2 Implementation Details

Following [13, 39], the image encoder f_θ consists of two convolutional blocks, each composed of a 5×5 convolutional layer followed by ReLU activation and a 2×2 max-pooling operation, progressively expanding the number of channels from the input to 32 and then to 64. A subsequent 1×1 convolutional layer decomposes the extracted features into visual primitives. For cross-modal alignment, a linear projection layer maps the visual primitives from the visual dimension d_v to the text embedding dimension d_t . The visual primitive space is initialized with orthogonal vectors to form primitive bases, and continuous local core visual primitives are quantized to their nearest neighbors to enable efficient knowledge sharing in the federated setting. For semantic text features, we employ the pretrained CLIP ViT-B-32 text encoder [28]. The balancing factor of the space loss in equation (6) is set to 0.3 following prior work [9, 35]. All experiments are conducted on a machine equipped with an NVIDIA

Table 1: Accuracy (%) and communication parameters of federated learning methods across diverse heterogeneous scenarios.

Scenarios	Pathological		Practical			Real-world		# of Comm. Params. ($\times 10^3$)
Datasets	Cifar10	MNIST	$\beta = 0.1$	Cifar100 $\beta = 0.3$	$\beta = 0.5$	OfficeCaltech10	DomainNet	
FedAvg [25] (AISTATS 2017)	60.16	97.25	33.12	35.06	35.50	61.31	56.83	582.026 _{142×}
FedProx [20] (MLSys 2020)	60.06	97.18	33.17	35.27	35.56	60.47	56.65	582.026 _{142×}
Ditto [19] (ICML 2021)	89.06	99.80	47.53	34.91	35.94	61.42	57.00	582.026 _{142×}
APPLE [24] (IJCAI 2022)	90.13	99.70	55.89	42.94	35.88	62.52	58.22	582.026 _{142×}
FedProto [32] (AAAI 2022)	89.92	99.85	52.43	37.37	31.67	61.31	57.62	<u>51.200</u> _{12.5×}
FedKD [37] (Nat Commun 2022)	90.39	99.86	53.42	39.56	33.80	61.89	58.40	77.019 _{18.8×}
FedRep [5] (ICML 2021)	91.21	99.86	51.21	37.33	29.82	63.31	<u>60.15</u>	576.896 _{140×}
FedRoD [4] (ICLR 2022)	91.05	99.87	50.50	38.97	34.20	61.73	58.52	582.026 _{142×}
FedAS [39] (CVPR 2024)	<u>91.78</u>	99.93	55.17	41.11	35.92	<u>66.98</u>	59.73	576.896 _{140×}
FedRIR [13] (WWW 2025)	91.64	99.93	<u>58.27</u>	<u>43.12</u>	<u>36.48</u>	66.14	60.02	576.896 _{140×}
FedVPL (Ours)	92.27 _{↑0.49}	<u>99.87</u> _{↓0.06}	60.74 _{↑2.47}	44.56 _{↑1.44}	37.95 _{↑1.47}	72.60 _{↑5.62}	64.63 _{↑4.48}	4.096 _{1.00×}

GeForce RTX 3090 GPU. Each task is repeated three times over 1000 communication rounds, using the SGD optimizer with a learning rate of 5×10^{-3} , a batch size of 10, and one epoch per local training round. We report the best accuracy achieved across runs.

5.3 Effectiveness

The experimental results in Table 1 demonstrate the superiority of FedVPL across diverse federated learning scenarios with ten state-of-the-art federated learning methods, ranging from explicit regularization to adaptive model partitioning. In pathological non-IID benchmarks like Cifar10, where clients are assigned disjoint subsets of classes, FedVPL demonstrates a clear advantage over existing methods, consistently achieving higher accuracy. This trend continues in more challenging practical non-IID settings, where severe label skew is introduced, with FedVPL again outperforming other approaches. In real-world scenarios, where data distributions are more complex and diverse, FedVPL still maintains a clear edge, outperforming traditional methods that typically struggle with domain shifts and label imbalances.

In addition to its performance advantages, FedVPL offers substantial communication efficiency. Specifically, it reduces communication overhead by 12.5× compared to FedProto and 98% compared to FedAvg, requiring only 4.096×10^3 parameters to be communicated. This substantial reduction in communication costs makes FedVPL a highly efficient solution for large-scale federated learning, where communication resources are often constrained.

The limitations of existing methods are evident from their performance gaps. Traditional FL approaches, such as FedAvg and FedProx, suffer significant accuracy degradation in non-IID scenarios, as their parameter-averaging mechanisms fail to reconcile conflicting local feature spaces. While regularization-based methods like Ditto improve personalization, their reliance on full-model synchronization renders them impractical for large-scale deployments, as their communication payload is 142× larger than that of



Figure 4: t-SNE visualization of features before classification on Cifar10, with each color represents an unique class.

FedVPL. Feature alignment methods, such as FedProto, reduce communication costs but sacrifice accuracy due to rigid class-level alignment, particularly in complex settings such as real-world scenarios. Model-splitting approaches such as FedAS and FedRIR address heterogeneity to some extent but still require significant communication resources, limiting scalability. For example, FedAS achieves 66.98% on DomainNet compared to FedVPL's 72.60%, while requiring 140× more communication parameters.

5.4 Feature Representation Diversity

FedVPL enhances the diversity and expressiveness of shared feature representations by decomposing class-level semantics into combinations of reusable visual primitives from the global space $\mathcal{S}_{\text{global}} \in \mathbb{R}^{K \times d_o}$. Unlike prototype-based methods such as FedProto, which represent each class by a single feature vector, FedVPL represents each class as a composition of M primitives selected from a shared set of K primitive bases. This compositional approach enables a combinatorial number of distinct class representations, fostering a much richer and more flexible semantic space. Consequently, FedVPL supports greater diversity across clients while maintaining a compact and efficient feature representation, which is both expressive and communicable.

The t-SNE visualization in Figure 4 further illustrates the diversity in feature representations. Samples from the same class are

Table 2: Test accuracy (%) for client amounts scalability.

	$N = 10$	$N = 20$	$N = 50$	$N = 100$
FedAvg	32.54	33.12	33.48	30.31
FedProx	32.65	33.17	33.47	30.38
Ditto	49.15	47.53	44.64	41.65
APPLE	52.53	55.89	49.64	33.99
FedProto	49.36	52.43	49.33	46.22
FedKD	54.59	53.42	50.38	46.42
FedRep	52.96	51.21	47.68	42.75
FedRoD	51.04	50.50	47.37	42.88
FedAS	57.75	55.17	48.11	39.07
FedRIR	57.58	58.27	53.63	50.31
FedVPL	59.08 _{↑1.33}	60.74 _{↑2.47}	56.28 _{↑2.65}	53.27 _{↑2.96}

distributed in a semantically structured manner, aligning with FedVPL’s core design principles. This dispersion reflects the model’s ability to capture fine-grained intra-class variations, such as sub-categories or contextual differences, which rigid prototype-based methods like FedProto cannot achieve. In contrast, FedAvg exhibits overlapping and fragmented clusters due to its inability to reconcile conflicting feature spaces, while FedProto enforces compactness, erasing intra-class diversity. FedVPL’s t-SNE patterns show a clear advantage: the reusability of visual primitives balances local adaptability and global consistency, enabling more diverse and accurate class representations.

5.5 Scalability

To evaluate the scalability of FedVPL, we simulate different federated learning scales by partitioning the Cifar100 dataset into 10, 20, 50, and 100 clients. As shown in Table 2, FedVPL consistently outperforms all baseline methods across different client configurations, demonstrating superior robustness to data fragmentation. When scaling from 10 to 100 clients, FedVPL maintains the highest accuracy in all scenarios, with only a moderate 5.81% decline (from 59.08% to 53.27%), significantly outperforming the second-best methods at each scale. Notably, the performance advantage over competitors grows as the number of clients increases: from a 1.33% lead over FedAS at 10 clients, to a 2.47% gap against FedRIR at 20 clients, and ultimately a 2.96% margin over FedKD at 100 clients. This expanding superiority contrasts sharply with other methods that exhibit significant degradation. For instance, APPLE suffers a dramatic 21.54% accuracy drop when scaling from 10 to 100 clients. The results highlight FedVPL’s unique capability to handle client heterogeneity and non-IID data challenges, likely attributable to its specialized design that effectively preserves critical knowledge during model aggregation while adapting to increasingly complex distributed learning environments. The widening performance gaps underscore FedVPL’s growing advantages in large-scale deployments, positioning it as a more scalable solution for real-world federated web systems.

Table 3: Test accuracy (%) with different participation ratios.

Algorithm	$\rho = 1.0$	$\rho \in [0.5, 1]$	$\rho \in [0.1, 1]$
FedAvg	60.16	56.37	55.78
FedProx	60.06	55.18	56.34
Ditto	89.06	89.34	89.29
APPLE	90.13	89.83	88.24
FedProto	89.92	89.53	88.44
FedKD	90.39	90.01	89.96
FedRep	91.21	90.56	90.61
FedRoD	91.05	88.33	87.76
FedAS	91.78	91.12	90.96
FedRIR	91.64	90.98	90.82
FedVPL	92.27 _{↑0.49}	91.43 _{↑0.31}	91.30 _{↑0.34}

5.6 Stability

To evaluate FedVPL’s stability under varying client participation scenarios, we simulate real-world dropout conditions by adjusting the client participation ratio ρ on the Cifar10 dataset. As shown in Table 3, FedVPL achieves state-of-the-art accuracy across all participation regimes while demonstrating exceptional robustness to participation instability. When all clients participate, FedVPL attains 92.27% accuracy, surpassing FedAS by 0.49%, thanks to its primitive-composition mechanism that enriches feature diversity. This advantage persists under challenging partial participation scenarios, maintaining 91.43% ($\rho \in [0.5, 1.0]$) and 91.30% ($\rho \in [0.1, 1.0]$) accuracy, surpassing the second best method FedAS by 0.31%-0.34% in accuracy. Crucially, while conventional methods like FedProto and FedRoD exhibit amplified sensitivity, FedVPL shows minimal degradation, benefiting from its visual primitive space sharing that preserves essential semantic components, regardless of participation patterns. The text-guided semantic alignment further ensures stable personalization by adaptively reconstructing discriminative features from shared primitives, even when only 10% of clients are involved. This contrasts with approaches like FedProto, where rigid feature alignment becomes vulnerable under participation uncertainty. The results validate that FedVPL’s primitive-composition paradigm fundamentally addresses the stability limitations of existing pFL methods, maintaining both generalization capability across clients and personalized adaptation quality.

5.7 Model-Agnostic Nature

Instead of sharing model parameters, FedVPL transmits the visual primitive space between the server and clients, allowing the server to aggregate visual primitives instead of gradients or parameters. This inherently eliminates the need for model consistency across clients. The results in Table 4 demonstrate FedVPL’s model-agnostic capability on OfficeCaltech10 dataset, as it outperforms FedProto even when client models differ, such as AlexNet, MobileNetV2, ResNet18, and Transformer. FedVPL achieves superior accuracy, confirming that its use of visual primitives instead of prototypes enhances performance across heterogeneous client models, further validating its model-agnostic nature.

Table 4: Test accuracy (%) with different client models.

Method	Client 1	Client 2	Client 3	Client 4	Avg.
	AlexNet	MobileNetV2	ResNet18	Transformer	
FedProto	65.00	34.52	72.50	68.92	52.44
FedVPL	70.83	39.86	82.50	69.46	65.66

Table 5: Privacy attack results on DomainNet. Lower is better for all metrics except MSE.

Method	DRA		MIA	
	PSNR ↓	MSE ↑	Acc ↓	AUC ↓
FedVPL (Primitive basis)	2.95 dB	0.5072	0.6000	0.7311
FedProto (Prototype)	5.21 dB	0.3010	0.8667	0.9467
FedAvg (Gradient)	3.39 dB	0.4585	0.6333	0.7267

5.8 Privacy Preservation

FedVPL offers robust privacy-preserving guarantees from three key perspectives. Firstly, by design, FedVPL communicates only a set of shared, low-dimensional primitive space bases, rather than client-specific prototypes, model parameters, or gradients. These bases are both class-agnostic and sample-independent. They are randomly initialized and then learned through the space loss function defined in equation (3) in the paper, ensuring that they are not directly derived from raw data and do not contain any sample-level semantics, which makes them inherently resistant to inversion.

Secondly, attackers cannot reconstruct raw data from primitive bases without access to the local models. Even under a stronger threat model where attackers are assumed to have such access, it remains infeasible to reconstruct raw data from these primitive bases alone. We conducted additional experiments on DomainNet to evaluate potential leakage on two primary attack types: Data Reconstruction Attack (DRA) and Membership Inference Attack (MIA). DRA attempts to reconstruct the original data from model parameters, while MIA tests whether an attacker can determine if a specific data point was included in the training set. The results in Table 5 demonstrate that FedVPL significantly outperforms existing methods, yielding lower reconstruction quality and reduced success rates in MIA, thus providing stronger empirical privacy protection.

Finally, FedVPL is fully compatible with formal privacy-preserving techniques such as differential privacy and secure aggregation, which can be easily incorporated to further enhance system reliability and compliance in practical deployments.

5.9 Ablation Study

To validate the effectiveness of FedVPL’s core components, we conduct ablation studies across three federated learning scenarios: Pathological (Cifar10), Practical (Cifar100), and Real-world (DomainNet). As shown in Table 6, the full FedVPL consistently outperforms all variants across all settings, demonstrating the complementary benefits of its visual primitive space sharing and text-guided semantic alignment mechanisms.

Table 6: Ablation study

Settings	Pathological	Practical	Real-world
Datasets	Cifar10	Cifar100	DomainNet
FedProto	89.92	52.43	57.62
FedProto w/ text-guided	90.51	54.39	60.13
FedVPL w/o text-guided	91.11	57.09	62.87
FedVPL w/o label mask	91.68	60.24	64.09
FedVPL	92.27	60.74	64.63

Removing the label masking mechanism (FedVPL w/o label mask) results in a slight performance drop across all scenarios, highlighting the importance of the label mask in preventing interference during the alignment process and ensuring that client-specific semantics are correctly captured. More notably, disabling text-guided alignment (FedVPL w/o text-guided) leads to a reduction in performance, though the model still performs well. This suggests that while visual primitives alone are effective, text-guided alignment provides a significant additional boost, particularly in Cifar100 and DomainNet, where accuracy improvements of 3.65% and 1.76% are observed, respectively.

To further isolate the impact of textual guidance and validate the effectiveness of the proposed visual primitives, we integrated text-guided alignment into FedProto (denoted as FedProto w/ text-guided). The results show that while FedProto benefits from the addition of text-guided alignment, it still lags behind FedVPL with its full text-guided alignment mechanism. These findings confirm that visual primitives themselves significantly enhance performance, while the inclusion of textual guidance provides an additional boost, further improving accuracy.

6 Conclusion

We propose FedVPL, a federated learning framework that combines visual primitive space sharing with text-guided semantic alignment, achieving significant improvements in effectiveness and communication efficiency. Extensive experiments across diverse scenarios demonstrate FedVPL’s superior performance, with accuracy gains of up to 5.62% and a 12.5× reduction in communication costs compared to state-of-the-art methods. Visual primitive sharing enhances feature diversity across clients, enabling flexible and efficient representation learning, while text-guided alignment ensures precise semantic grounding without additional communication overhead. Furthermore, FedVPL shows strong scalability and stability across diverse client configurations, making it particularly well-suited for large-scale federated learning with heterogeneous data and dynamic client participation.

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A The Learning Procedure

Algorithm 1: The learning procedure of FedVPL

Input: N clients $\{C^i\}_{i=1}^N$ with their local data $\{\mathcal{D}^i\}_{i=1}^N$; pretrained class-text embeddings $\mathbf{T}$; initial global visual primitive space $\mathcal{S}_{\text{global}}$; initial weights of image encoder f_θ , visual primitive projector $p_{\mu,\Sigma}$, and visual primitive fusion layer ϕ ; client joining ratio ρ ; total communication rounds R .

Output: Trained parameters of $\mathcal{S}_{\text{global}}$, f_θ , $p_{\mu,\Sigma}$, ϕ .

- 1 Each client i initializes its local models f_θ , $p_{\mu,\Sigma}$, ϕ , and local visual primitive space $\mathcal{S}_{\text{local}}^i$;
- 2 **for** iteration $r = 0$ to $R - 1$ **do**
- 3 Server samples a subset of clients $\{C^i\}_{i=1}^{\rho \times N}$;
- 4 Server broadcasts $\mathcal{S}_{\text{global}}$ to all selected clients.
- 5 **foreach** client C^i in $\{C^i\}_{i=1}^{\rho \times N}$ **in parallel do**
- 6 // Visual Primitive Space Sharing
- 6 Extract latent features z of the local data with f_θ and project them into $\mathbf{v}_\mu$ and $\mathbf{v}_\Sigma$ via $p_{\mu,\Sigma}$ by Eq. (4);
- 7 Quantize $\mathbf{v}_\mu$ from $\mathcal{S}_{\text{local}}^i$ to obtain $\mathbf{v}_q$ using Eq. (5);
- 8 Concatenate $\mathbf{v}_q$ and $\mathbf{v}_\Sigma$ and adjust the dimension via ϕ following Eq. (7).
- 9 // Text-Guided Semantic Alignment
- 9 Mask each element t_c in $\mathbf{T}$ using client-specific class indices I^i to obtain $\mathbf{T}^i$ by Eq. (9);
- 10 Compute all $\mathcal{S}_{m,c}$ and sum them along the visual primitive dimension to get class-wise scores $\mathbf{S}$ by Eq. (12);
- 11 Obtain the prediction by applying Softmax to $\mathbf{S}$ using Eq. (13);
- 12 Update $\mathcal{S}_{\text{global}}$, f_θ , $p_{\mu,\Sigma}$, ϕ by minimizing $\mathcal{L}_{\text{total}}$.
- 13 // Server Aggregation
- 13 Server collects $\mathcal{S}_{\text{local}}^i$ from all selected clients;
- 14 Server aggregates $\mathcal{S}_{\text{global}}$ by Eq. (8).
- 15 **return** Trained model parameters $\{f_\theta, p_{\mu,\Sigma}, \phi\}_{i=1}^N$ and global visual primitive space $\mathcal{S}_{\text{global}}$.

B Hyper-parameters about Primitive Space

The trade-off between the number of primitive bases K and their dimensionality d_v plays a key role in balancing representation capacity, communication efficiency, and generalization in FedVPL. As shown in Figure 5, our experiments across varying configurations highlight the relationship between these two factors. For instance, when $K = 16$ and $d_v = 256$, we observe a balanced performance, with an accuracy of 60.74%, which is one of the highest among the different configurations. Increasing the number of primitive bases to $K = 32$ and $d_v = 256$ results in a slight improvement to 60.76%, but the gains in accuracy are marginal.

The analysis of different dimensionalities further reveals diminishing returns with increasing d_v . For example, with $K = 16$ and $d_v = 512$, the accuracy only increases to 60.76%, which is nearly

identical to the result with $K = 16$ and $d_v = 256$ (60.74%), even though the number of parameters doubles. Similarly, for configurations with higher base values, such as $K = 128$ and $d_v = 256$, the accuracy reaches 60.66%, and for $K = 256$ with $d_v = 256$, the accuracy is 60.60%, showing that larger values of K and d_v do not consistently provide significant improvements in performance. In fact, as K increases, the accuracy tends to stabilize around 60.5%, suggesting that beyond a certain point, the communication overhead outweighs the potential benefits in accuracy.

These findings suggest that $K = 16$ and $d_v = 256$ provides the best trade-off between discriminative power and communication efficiency, while further increases in K or d_v offer diminishing returns in terms of accuracy but significantly increase the communication cost.

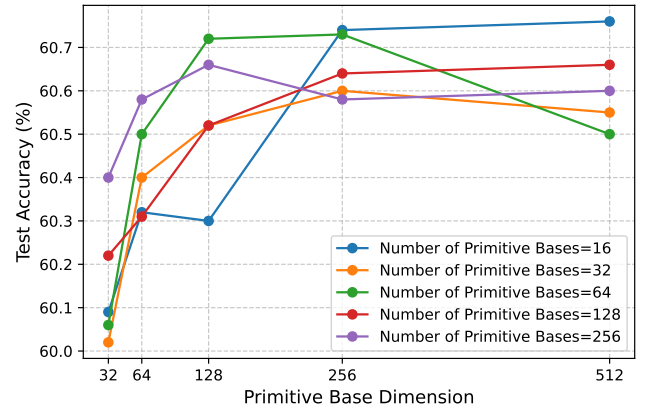


Figure 5: Test Accuracy on Cifar100 ($\beta = 0.1$) under different number of primitive bases and primitive base dimensions

C Analysis of Training Dynamics

The training loss and test accuracy trajectories presented in Figure 6 demonstrate distinct behavioral patterns across various personalized federated learning (pFL) methods under heterogeneous settings. These results underscore the effectiveness of FedVPL in achieving a favorable trade-off between stability and generalization.

Although FedVPL does not attain the most rapid initial loss reduction, its convergence curve is notably stable, avoiding the sharp oscillations characteristic of several baseline methods. In contrast, regularization-based approaches such as FedAvg, FedProx, and Ditto exhibit nearly identical loss trajectories, suggesting a limited capacity to adapt to data heterogeneity. While knowledge distillation techniques (e.g., FedKD) and feature alignment strategies (e.g., FedProto) display more stable convergence, but this stability comes at the expense of flexibility in feature representation. For instance, FedProto's use of class-level prototype alignment imposes excessive rigidity in feature space exploration, leading to suboptimal convergence outcomes.

Moreover, the divergence between training loss and test accuracy trajectories highlights the overfitting tendencies of several methods. FedAvg and FedProx, for example, reach a plateau in training loss relatively quickly, yet suffer substantial degradation in test

accuracy beyond 100 epochs, indicating overfitting to local data biases rather than learning transferable features. Model-splitting approaches such as FedRep and FedAS achieve promising performance initially, but encounter significant accuracy fluctuations. These failures can be attributed to the fragmentation of model architectures; the decoupled global and local components struggle to preserve semantic consistency during extended training. In contrast, FedVPL attains the highest final test accuracy with minimal variance post-convergence, demonstrating that its primitive-composition mechanism effectively harmonizes generalization and personalization throughout the learning process.

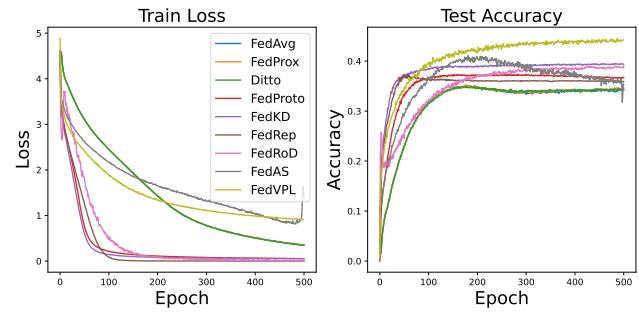


Figure 6: Training loss and test accuracy curves of compared methods on Cifar100 classification task.